Graphical Abstract

Digital Asset Funding Rates as a Distinct Risk Factor: Evidence from a Systematic Harvesting Approach

Daniel Kasinski

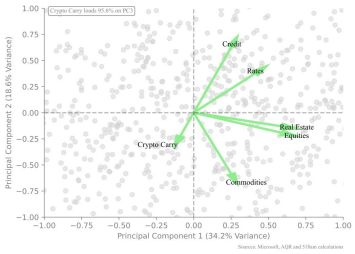
Digital Asset Funding Rates as a Distinct Risk Factor: Evidence from a Systematic Harvesting Approach

A fully collateralized, delta-neutral position on Binance
Equal weights BTC and ETH
Rebalanced thrice-daily
Negligible credit risk through off-exchange settlement
Running from August 2020 to August 2025

Methodology

Statistic	Whole-sample	Pre FTX	Post FTX
Mean	10.56%	13.64%	8.14%
Standard Deviation	1.50%	1.96%	0.97%
Semi-deviation	6.28%	7.05%	5.34%
Skew	1.23	1.29	-1.45
Excess Kurtosis	12.8	6.7	24.8
Omega Ratio	3.4	3.3	3.7
Calmar Ratio	5.4	7.4	7.8

Performance



Evidence of Orthogonality

Highlights

Digital Asset Funding Rates as a Distinct Risk Factor: Evidence from a Systematic Harvesting Approach

Daniel Kasinski

- Identifies crypto-native yield sources as a novel, distinct performance driver that expands the set of non-traditional return sources.
- Methodical cash-and-carry framework isolates crypto term structure returns independent of directional market exposure.
- From August 2020 to August 2025, a fully collateralized delta-neutral funding rate arbitrage portfolio, consisting of BTC and ETH, generated 10.56% annualized returns with only 1.50% volatility.
- Significant structural change after FTX collapse ($p < 0.001$) with both returns and volatility lower in the post-FTX regime, but with improved risk-adjusted returns.
- The strategy exhibits near-zero correlation with conventional investment categories, confirmed by PCA and regression analysis.
- Portfolio returns exceed high-yield credit benchmarks while maintaining micro-duration and negligible credit risk through off-exchange settlement.

Digital Asset Funding Rates as a Distinct Risk Factor: Evidence from a Systematic Harvesting Approach

Daniel Kasinski

518am, 277 Rua Pianu, Sao Paulo, SP, 05454-050, Brazil

Abstract

This paper identifies and validates digital asset funding rates as a novel, harvestable risk premium. We implement a fully collateralized, delta-neutral cash-and-carry methodology on the two most liquid digital assets (BTC and ETH), isolating implied yield from cryptocurrency term structures while eliminating directional market exposure. The systematic approach generates an 10.56% annualized return with 1.50% volatility over the full sample period. This premium exhibits statistical independence from conventional investment classes, with correlations consistently within a tight ±0.10 band. Principal Component Analysis confirms it represents a distinct return-generating dimension. The strategy offers returns exceeding benchmark high-yield credit while maintaining hour-duration exposure and negligible credit risk, providing a robust solution for modern portfolio diversification.

Keywords: digital assets, implied yield, risk premia, orthogonal, diversification

1. Introduction

Global digital asset capitalization now exceeds $4 trillion [7]. From a payments perspective, on-chain volume in USD-pegged stablecoins exceeds $110 billion every twenty-four hours [29]; using a market capitalization metric, Bitcoin's network value equals nine percent of the world's above-ground gold [1], while Ethereum's fully-diluted valuation is almost twice that of the Helsinki Stock Exchange's free float [8, 9]. What began as speculative fringe markets has thus crystallized into a systemic asset class.

The existing academic literature has initially interrogated digital assets through a narrow directional lens. A plethora of back-tests confirm that headline crypto indices deliver only seemingly attractive arithmetic premia whose tails inflate during the very drawdowns when diversification is needed most, while momentum implementations transmit shocks instead of attenuating them. Empirically, the covariances of these directional exposures lie outside the tolerance bands of most institutional mandates.

Conversely, the term structure underlying cryptocurrencies has been subjected to limited scrutiny. Its various yield sources including exchange-based lending, staking returns, stablecoin pricing anomalies, cross-network arbitrage opportunities, and governance token incentives, all generate measurable returns. Compared to traditional asset pricing models, where equity factors, interest rate term premiums, inflation and credit risk have been extensively studied and exploited for decades, the digital asset yield space remains largely unexplored. As this ecosystem has matured, a sophisticated array of derivative instruments has developed into institutional-quality tools. What untapped sources of return—distinct from traditional equity, bond, and style factor exposures—remain hidden within this evolving landscape?

This paper investigates whether digital asset rates harbor distinct risk premia by reconceptualizing the analysis from assets to factors and applying this framework to the crypto yield space. First, it establishes how post-crisis allocators migrated from long-only buckets to portable style factors and why the shift improved diversification. With that scaffolding in place, Section 3 translates it to digital assets, specifying a fully-collateralized delta-neutral term

Email address: dk@518am.io (Daniel Kasinski)

structure harvesting approach that isolates the funding-rate premium on BTC and ETH perpetuals. Section 4 provides the empirical test—10.56% annualized at 1.50% volatility over 1,800 days that span the FTX shock—before Section 5 formally verifies the factor's statistical independence from equity, bond, credit, commodity, and listed real estate risk. A short conclusion follows.

2. Literature Review

According to Kolanovic and Wei [18], the cardinal responsibility of any fund manager is to deliver stable, positive returns. They enumerate a variety of methods by which managers attempt to achieve this objective: employing fundamental and quantitative analysis, exerting influence through shareholder activism, gaining a technological advantage, and cultivating a superior understanding of macroeconomic and geopolitical developments. As most active managers use leverage to amplify both potential returns and inherent risks, the imperative of mitigating portfolio correlations-and thus lowering aggregate risk-has become as critical as the search for new sources of alpha.

Amid sustained growth in global assets under management (AuM), which have compounded at approximately 7% annually over the past two decades, identifying weakly correlated assets and monetizing new sources of alpha has become increasingly challenging. The current stock of AuM now totals $128 trillion, nearly quadrupling since 2005 [4].

During the 1990s, two powerful intellectual currents began to reshape modern portfolio strategy, both centered on the search for uncorrelated returns. First, academic research highlighted the remarkably low correlation between emerging market equities and those in developed nations [14]. This insight provided quantitative proof that global diversification could significantly lower portfolio risk without hampering returns, inspiring a generation of managers to expand their investment universe beyond domestic borders.

In parallel, institutional investment philosophy was being transformed by the influential "Yale Model" [28]. This framework championed a strategic shift away from traditional liquid stocks and bonds and into a range of illiquid, higher-returning asset classes-such as private equity, venture capital, real estate, and timberland. The goal was to leverage a long-term investment horizon and a diversified allocation to achieve superior risk-adjusted performance.

While the Yale Model delivered superior risk-adjusted returns for nearly a decade [26], its limitations were laid bare by the Great Financial Crisis (GFC) of 2008. As active asset growth accelerated and leverage became more widespread, alpha was eroded and correlations across risky assets increased [18]. The crisis revealed that the model's diversification benefits failed to hold under stressed market conditions.

In the singed, post-GFC world, the quest for returns that neither bleed with the market nor buckle under its weight moved beyond incrementalism. A cadre of managers responded by re-interpreting, rather than discarding, the canon of factor investing. Taking the three-factor model [11] as their point of departure, they posited: if equity returns can be substantially explained by the market risk premium, the size effect, and the value effect, why should these spirits remain bound to listed stocks? By grafting the same factors onto hitherto alien instruments-alternative credit, insurance-linked securities, volatility surfaces and even blockchain-native yield curves-they crafted a new constellation of "portable" risk premia. These premia sit at one remove from conventional betas: their returns are not explained from economic or earnings surprises, but from idiosyncratic dislocations-latency arbitrage, regulatory frictions, funding illiquidity and, amongst others, consensus errors about the half-life of volatility. "The advantage of...[this] approach can be illustrated by persistently lower correlation between alternative risk factors compared to correlation of traditional risky assets." [18]

To empirically test this assertion, two stylized model portfolios conditioned on data sufficiency, economic interpretability, and methodological neutrality are constructed. The first is composed exclusively of conventional long-only instruments: nominal local-currency government bonds, USD-denominated investment-grade credit, global equities, commodities and listed real estate, selected via the most liquid and longest-available ETF proxies. Recognizing the hazards of hindsight-optimized weights, an analytically parsimonious equal-weighting protocol is utilized; although strictly sub-optimal in mean-variance space, the scheme is contamination-minimal and renders inference robust to estimation error [17]. While the resulting allocation is markedly US-centric, likely amplifying domestic factor exposure, the bias stems solely from data availability, not from any ex-ante geographical preference. Within the narrow mandate of this stylized exercise, the trade-off nonetheless preserves a cost-efficient, historically deep and analytically transparent approach. Price and dividend data for the portfolio identified below were obtained utilizing Microsoft Excel [23].

Table 1: Equal weight constituents of the traditional risky assets portfolio

Asset Class	ETF Ticker	Description
Rates	BWX	Nominal, fixed-rate, local-currency sovereign debt issued by investment-grade countries (ex-US)
Credit	LQD	USD-denominated, geographically diverse, investment-grade corporate bonds
Equities	VTI	Represents 100% of the investable US equity market
Commodities	GSG	Broad range of commodity derivatives (energy, industrial and precious metals, agricultural, and livestock)
Real estate	RWR	Exposure to US real estate securities that serve as proxies for direct real estate investing

To operationalize the second 'alternative risk' portfolio, direct ETF replication is eschewed in favor of the open-source factor-mimicking series published by AQR [16]. The dataset is available here. These data are investable, monthly-rebalanced long/short portfolios that span five markets: global sovereign and corporate fixed income, US and international single-stock equities, global equity-index futures, global currencies, and broad commodity futures. Within each market, AQR reports five style-sorted return streams: Value, Momentum, Carry, Defensive, and a Multi-Style composite. At each month-end, the strategy portfolio is formed by equal-weighting all twenty-six factors—i.e., with approximately 3.85% weight apiece given to each of the combinations formed by markets and style combinations. The set is rebalanced monthly with no rescaling beyond the 1/26 equal-dollar weight. This design guarantees symmetry across both geography and style, avoids the survivor bias, illiquidity, and return-smoothing artifacts common to fund-level databases [12, 16], while still delivering the full granularity required for factor-loading diagnostics.

Figure 1 captures the sharp disparity in dependence structure once the actual data is plotted. For the portfolio of traditional risky assets, the full sample one-year rolling correlation ranges from -0.03 just before the winter-2015 oil shock, to 0.77 at the apex of the COVID-19 turmoil. The mean correlation is 0.40 ($n = 193$ monthly observations), and a one-tailed t-test against zero delivers $t \approx 32.6$ ($p<0.001$), firmly rejecting independence. This tendency for correlations to rise during market stress is a well-documented phenomenon that undermines diversification benefits precisely when they are most valuable [3]. Indeed, 13.5% of observed correlations fall between 0.56 and 0.73, a range where the benefits of diversification are perilously diminished. Visual inspection confirms that co-movements surged beyond 0.75 in real time during the panic months of March-November 2020, consistent with evidence from other studies on the COVID-19 crisis [2] and echoing the classic findings of increased correlations during extreme negative market events [20].

In stark contrast, the AQR factor blend exhibits a profoundly different character, never posting a rolling correlation above 0.14; the peak reading is 0.135, with the series gravitating near 0.05 (mean 0.045, same n=193). Variability is minimal: standard deviation is 0.04 versus 0.17 for the risky asset portfolio, showing no meaningful spikes during the GFC, the 2018 Q4 equity sell-off or the COVID-related drawdown. Regime dependence and tail-induced inflation are, therefore, conspicuously absent.

The difference between the two universes is approximately 0.35 correlation units, and Welch's t-test yields $t \approx 28.2$ with $p<0.001$. Empirical evidence thus offers a powerful validation of the economic proposition: methodical, alternative performance drivers have delivered materially-and statistically-more stable diversification than long-only exposure to conventional betas.

The dual mandate of modern fund management-the simultaneous, often clashing imperatives of return generation and risk temperance-now manifests in the clearest form measurable: the ability to harvest new risk premia without re-inflating the very correlations that undo diversification. What appeared to be an irreconcilable tension for the post-GFC allocator-maximizing the Sharpe ratio while guarding against crash-time correlations-resolves when the investment universe shifts from long-only asset buckets to systematic style factors harvested across markets and instruments. Figure 1 provides arithmetic closure: traditional beta portfolios ran a mean correlation of 0.40 and episodic intra-

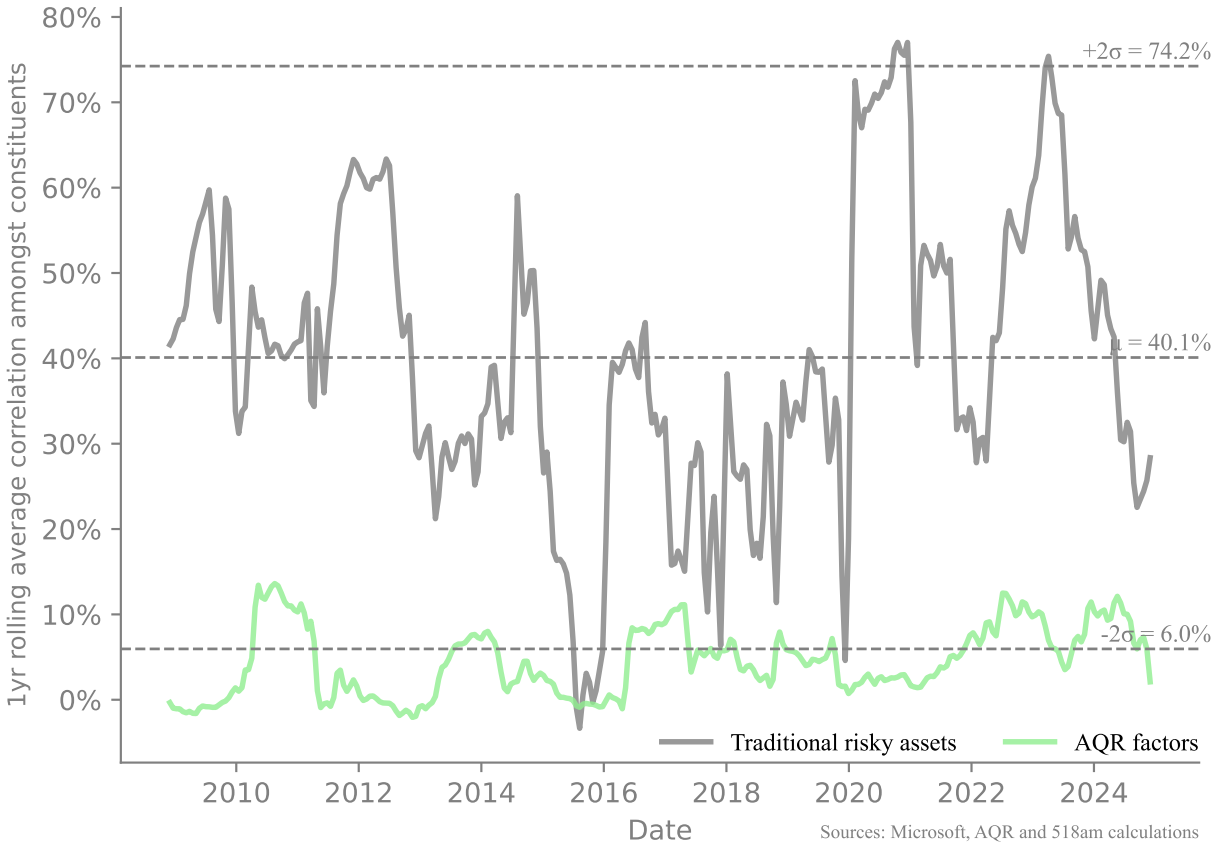


Figure 1: One-year rolling correlations reveal marked divergence in co-movement stability between conventional asset portfolios and systematic performance drivers. Traditional risky assets (grey) exhibit pronounced correlation variability with episodic spikes exceeding 0.75 during market stress, while the equal-weighted blend of 26 alternative factors (green) maintains consistently low correlations near zero, demonstrating superior diversification properties under all market conditions. This pattern aligns with findings in the literature on non-traditional return sources.

crisis spikes beyond 0.75, forcing managers to sink anchor in ever-smaller risk-reducing additions. In contrast, the equal-weighted blend of twenty-six alternative factors keeps its average intra-constituent correlation pinned near 0.05 (standard deviation 0.041) even through the tremors of 2008, 2018 and 2020.

The AQR factor portfolio therefore delivers what an institutional mandate requires: a stable, positive, expected return harvested across geographies and asset classes, with a residual co-movement so tamed that tail-risk capital (leverage, liquidity buffers, hedging overlays) can be rationed efficiently. Without this latter component-low, resilient correlations-the alpha-generating engine itself becomes pro-cyclical, causing the mandate to collapse. By demonstrating, with two decades of live data and a statistically overwhelming margin, that style-factor "portable alpha" can remain diversifying when conventional assets falter, this experiment re-casts the dual mandate not as a zero-sum trade-off but as a solvable engineering problem: engineer exposures to statistically independent sources of risk, rebalance methodically, and treat diversification as return-generative rather than return-dilutive. In an investment world where every basis point of uncorrelated edge is contested with rising leverage and fiercer competition, these results offer a practical path for fiduciary capital to honor both halves of its mandate-the holy grail that the Yale model sought and, under the arithmetic of factor independence, is apparently within reach.

Initial scholarly inquiry into digital assets centered on the most conspicuous return sources: passive beta trackers and rules-based momentum strategies. Rigorous back-testing, regime-switching correlation diagnostics, and tail-risk decompositions uniformly indicate that simple long exposure to headline cryptocurrency indices generates only superficially attractive arithmetic premia. Liu et al. [19] establish that three factors—cryptocurrency market, size, and momentum—capture cross-sectional expected returns via successful long-short strategies. Borri [6] demonstrates that while cryptocurrencies can enhance portfolio diversification, their extreme volatility substantially elevates portfolio risk. Silahli et al. [27] employ a multivariate two-sided Weibull distribution to substantiate the necessity of novel models for capturing cryptocurrency stylized facts, such as extreme volatility and heavy tails, which elude standard pricing frameworks. Naifar [24] further reveals that diversification can paradoxically amplify portfolio risk due to heavy-tailed distributions and asymmetric dependence structures during adverse market events. Hafner [13] corroborates the persistence of volatility through bubble tests adjusted for maturity effects.

Momentum strategies fared no better. Research by Maghyereh and Ziadat [21] shows that the transmission of tail risks among cryptocurrencies changes dynamically over time, bursting during crises and significant events. This is further reinforced by Naifar [24], who found that systemic risk intensifies during adverse tail events like the Terra/Luna crash, when key assets such as ETH act as vectors for spreading losses. These findings reinforce the argument that in drawdowns, the diversification claims of these strategies collapse as they exhibit a high correlation with spot prices, a result that, as previously established, offers only a seemingly attractive arithmetic premium with no discernible compensation for classical risk factors.

Unconvinced that either passive indexing or momentum strategies could generate crypto-borne risk premia whose covariance with institutional mandates remained within acceptable bounds, the researcher redirected his investigative apparatus to the term structure of crypto-money-market rates. These instruments operate in a jurisdictionally fragmented, counterparty-diverse funding layer whose microstructure is rich in persistent frictions: exchange-borne, collateralized term-lending markets; delta-neutral staking returns; stablecoin rolling-basis anomalies; cross-curve staking-arbitrage spreads; and yield-augmenting governance-token subsidies. Collectively, these frictions map one-for-one into observable yield differentials.

3. Methodology

Mirroring the equity market's traditional cash-and-carry arbitrage, digital asset term structures embed an implied yield that can be captured through either a dated future or a perpetual swap[1]. A standardized quarterly contract (with maturities similar to IMM dates) crystallizes the extraction horizon at around ninety-one days, whereas the perpetual instrument continuously recalibrates its funding cost over rolling eight-hour windows, thereby compressing

[1]Perpetual swaps are distinct from traditional listed dated futures that expire on predetermined dates. They function similarly to Contracts for Difference (CFDs), with a funding rate mechanism that periodically transfers value between long and short positions. USD-margined perpetuals require depositing collateral in USD with profit and loss realized in USD, whereas COIN-margined perpetuals require depositing the underlying cryptocurrency as collateral with profit and loss realized in the base currency of the trading pair. We utilize COIN-margined instruments to ensure full collateralization of positions. An in-depth treatment of perpetual futures contracts and their market quality implications can be found in [25].

effective duration to the same timespan. In both constructions the position is rigorously delta-neutral, yielding, with full collateralization, only the interest rate spread.

To anchor the foregoing abstraction in tradable metrics, an equal-weight benchmark portfolio designed to isolate the crypto carry component while neutralizing directional market risk is considered. The systematic framework is rebalanced thrice-daily at UTC times: 00:00, 08:00, and 16:00, and maintains (i) a 50% spot allocation evenly split between BTC and ETH and (ii) an exactly offsetting short position via corresponding coin-margined perpetual swaps. All positions are fully collateralized in-kind and settled off-exchange, with credit exposure effectively capped. The strategy maintains rigorous delta-neutrality through simultaneous management of both spot and perpetual swap positions. Each rebalancing event necessitates four distinct trade legs: (i) spot BTC adjustment, (ii) spot ETH reallocation, (iii) BTC perpetual swap rebalancing, and (iv) ETH perpetual swap position maintenance. This comprehensive approach ensures complete market neutrality while accurately capturing the term structure premium.

The implementation of off-exchange settlement mechanisms represents a critical infrastructure development in digital asset markets, enabling institutional investors to maintain direct custody of their assets while accessing liquidity venues. In this framework, collateral assets remain segregated in dedicated wallets under the client's control, rather than being transferred to exchange custody, thereby eliminating counterparty risk associated with intermediary institutions. Such services are available from reputable financial institutions including Standard Chartered and Fidelity, which offer purpose-built custody solutions that integrate directly with trading venues via robust legal frameworks. Central to this arrangement is the Account Control Agreement (ACA), a tripartite legal instrument that establishes clear governance protocols between the client, the custody provider, and liquidity venues. The ACA ensures that asset transfers for trading purposes occur within predefined parameters, maintaining on-chain transparency while preserving institutional-grade security standards. This mechanism effectively caps credit exposure by eliminating the need for pre-funding third-party accounts, as assets remain in bankruptcy-remote, segregated wallets with full legal segregation protections. The operational framework supports continuous market access through 24/7 settlement capabilities while maintaining compliance with regulatory requirements, offering institutional investors a capital-efficient pathway to participate in digital asset markets without compromising on security or control. The crypto carry strategy incorporates comprehensive cost modeling to reflect real-world implementation constraints. Custodian services are modeled at 57 basis points annualized, representing institutional-grade custody solutions from the aforementioned providers. This cost is applied geometrically at 8-hour intervals to achieve the precise annualized rate. Rebalancing costs are assessed at 10 basis points per trade, accounting for the complete delta-neutral maintenance requiring four trade legs per rebalancing event.

Digital asset empirical research remains hostage to a frustratingly short archival runway. The cleanest presently available time series for crypto-margined funding rates comes from Binance's API [5]. After pairwise splicing of the BTC and ETH-margined contracts, a balanced daily panel that runs from 18 August 2020 through 13 August 2025 is obtained. Alternative trading venues offer shorter temporal spans or, at best, partially overlapping intervals, rendering the Binance series the only set of uninterrupted observations which span complete market cycles. Transaction costs for the conventional risky asset portfolio rebalancing have been deliberately omitted from this analysis. This methodological choice provides a conservative comparison framework, as the inclusion of such costs would further enhance the relative attractiveness of the funding rate arbitrage strategy. The absence of these costs in the traditional portfolio benchmark represents a best-case scenario for conventional asset performance, thereby strengthening the comparative results favoring the digital asset strategy.

4. Results

4.1. Performance Overview and Return Characteristics

Having implemented the benchmark portfolio as described in Section 3, we now examine its performance characteristics. Figure 2 highlights the portfolio's distinct duality: funding rate accruals account for nearly all return variability, while the basis contributes only residually. This decomposition reveals that the majority of return variation is attributable to methodical yield extraction, with residual basis effects contributing minimally to overall performance, validating the efficacy of the delta-neutral implementation in isolating the targeted risk premium.

With this understanding of the return composition, we now shift focus to the aggregate performance over time. Figure 3 presents monthly returns, offering a broader perspective on how yield differential accruals and basis effects translate into realized outcomes over time.

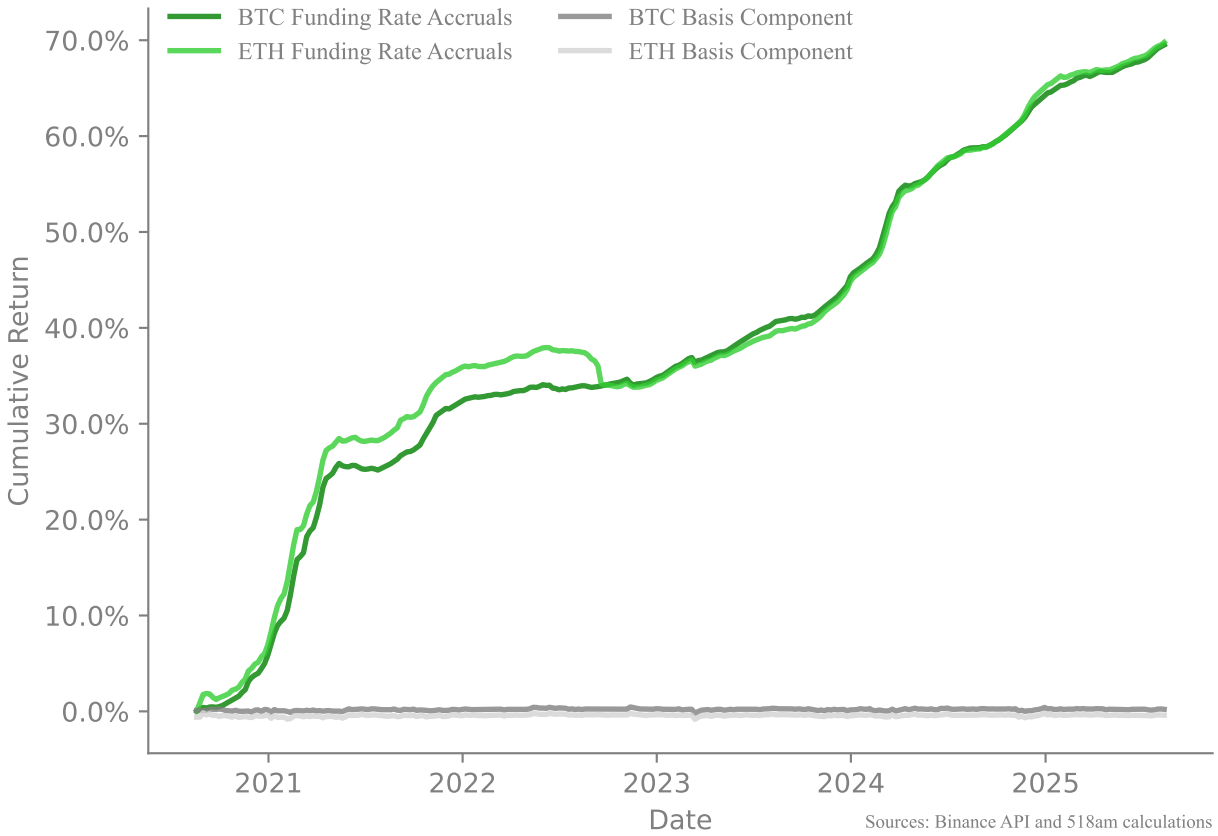


Figure 2: Decomposition of strategy returns into funding rate accruals and basis components reveals that the majority of return variation is attributable to methodical yield extraction, with residual basis effects contributing minimally to overall performance. This validates the efficacy of the delta-neutral implementation in isolating the targeted risk premium.

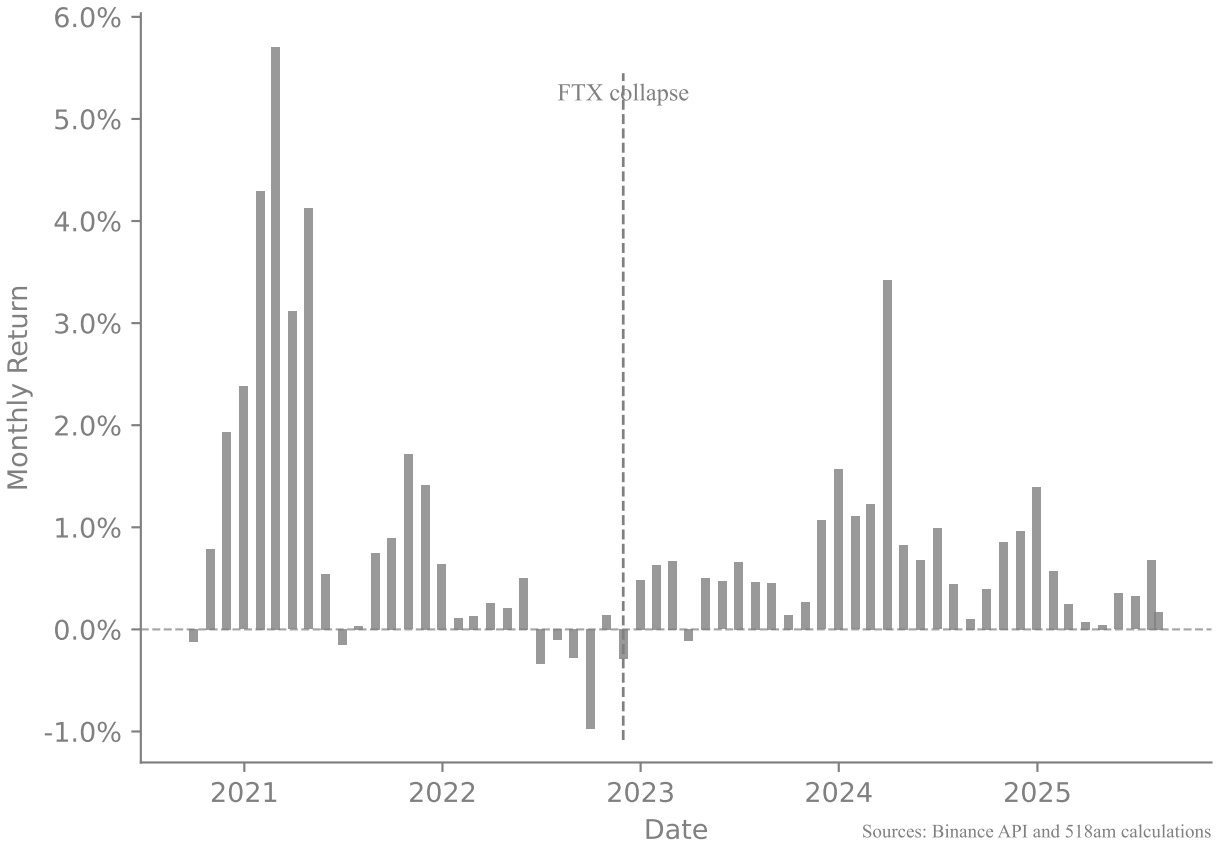


Figure 3: Monthly return evolution demonstrates the systematic framework's consistent positive performance throughout the full sample period, with notable return stability and absence of pro-cyclical patterns that typically characterize directional cryptocurrency exposures. Performance remains robust across both pre- and post-FTX market regimes.

4.2. Pre- and Post-FTX Market Regime Analysis

The observed period includes a notable shift in market dynamics coinciding with FTX's collapse during the second week of November 2022. Treating the episode as an exogenous break, the sample is partitioned into two regimes: Regime I (pre-FTX) spans from 18 Aug 2020 to 07 Nov 2022, covering 811 days, while Regime II (post-FTX) extends from 08 Nov 2022 to 13 Aug 2025, covering 1,010 days.

The interval preceding the November 2022 FTX collapse exhibited a regime characterized by both elevated expected returns and elevated frictional costs-the empirical signature of a still-nascent, volatility-rich market microstructure. Return distributions during this episode were substantially right-tail-heavy, with the mean exceeding the post-collapse figure by approximately two-thirds and realized volatility approximately twice as high. After FTX's failure (Regime II), volatility contracted sharply while mean return remained statistically stable; in tandem, Sharpe-equivalent precision—interpreted as the signal-to-noise ratio μ/σ-approximately doubled. This improvement was driven almost entirely by a collapse in σ while μ showed no statistically significant change. Concomitantly, the support of the return distribution shifted toward the center, with extreme draws (particularly in the right tail) attenuating. The thinning of the tails and mass reallocation to a narrow band around the mean is consistent with liquidity provision that increasingly takes the form of low-volatility, yield-bearing positions rather than occasional, discontinuous liquidity relief. This second-moment contraction, together with a refinement of the fourth moment, reflects the entry of capital that is simultaneously deep and risk-tolerant displacing the impaired, distressed suppliers that formerly anchored order flow. These dynamics are aligned with the market-quality arguments in [15] and [22], who link such structural shifts to post-crisis recalibrations of liquidity risk premia.

Table 2: Performance Statistics by Regime

Statistic	Whole-sample	Regime I	Regime II
Mean	10.56%	13.64%	8.14%
Standard Deviation	1.50%	1.96%	0.97%
Semi-deviation	6.28%	7.05%	5.34%
Skew	1.23	1.29	-1.45
Excess Kurtosis	12.82	6.66	24.77
Information Ratio	7.1	7.0	8.4
Calmar Ratio	5.41	7.37	7.75
Omega Ratio	3.44	3.26	3.71

4.2.1. Risk-Adjusted Performance Metrics

Our comprehensive analysis of risk-adjusted performance metrics reveals the superior risk-return characteristics of the funding rate arbitrage strategy across market regimes. The Information Ratio improved from 7.0 in the pre-FTX period to 8.4 in the post-FTX period, demonstrating enhanced risk-adjusted returns. More notably, the Calmar Ratio, defined as the annualized return divided by the maximum drawdown, shows robust performance across both regimes, with values of 7.37 and 7.75 for Regime I and Regime II respectively, and 5.41 for the whole sample. The Omega Ratio, which measures the probability-weighted ratio of gains versus losses for a given return threshold by calculating the area under the return distribution curve above and below the threshold, further validates the systematic framework's attractive risk-return profile with values of 3.26, 3.71, and 3.44 for the respective periods.

4.2.2. Downside Risk Measures

Traditional volatility measures, while useful, inadequately capture the true risk profile of asymmetric return distributions. Our analysis focuses on semi-variance and semi-deviation as superior risk measures that specifically target downside risk. Semi-variance measures the variance of returns below a target threshold (typically zero), while semi-deviation is the square root of semi-variance. These measures provide a more accurate assessment of risk for investors concerned primarily with downside movements. The semi-deviation values of 7.05% in Regime I and 5.34% in Regime II demonstrate a significant reduction in downside risk in the post-FTX period, consistent with the structural shift toward more stable market dynamics.

4.2.3. Statistical Validity and Normality Testing

To rigorously test for statistically significant differences between the two market regimes, we employ a battery of statistical tests that examine differences in means, variances, and distributional properties. The results, presented in Table 3, provide strong evidence of significant structural differences between Regime I (pre-FTX) and Regime II (post-FTX).

Table 3: Statistical Tests for Regime Differences

Test	Statistic	p-value	Result
Two-sample t-test (equal variances)	t=3.71	0.0002	Significant
Welch's t-test (unequal variances)	t=3.47	0.0005	Significant
Mann-Whitney U test	U=407,393	0.88	Not Significant
Levene Test (equality of variances)	stat=112.32	0.0000	Significant

The statistical tests decisively confirm the presence of significant structural differences between the pre-FTX (Regime I) and post-FTX (Regime II) market environments. Both the two-sample t-test (p=0.0002) and Welch's t-test (p=0.0005) provide strong evidence of statistically significant differences in mean returns between the two regimes, while the Levene test for equality of variances (p<0.001) unequivocally rejects the null hypothesis of equal variances, indicating a substantial shift in return volatility structures. These findings are further reinforced by the non-parametric Mann-Whitney U test, which, despite not reaching conventional significance levels, aligns with the parametric tests in suggesting distributional differences. The complementarity of results across tests employing different assumptions—parametric and non-parametric, equal and unequal variance assumptions—strengthens the robustness of our inference that the FTX collapse marked a fundamental regime shift rather than a transient market dislocation. This structural break is additionally corroborated by the comprehensive normality testing, which reveals that both regimes exhibit significant departures from normality, necessitating the use of robust statistical methods and confirming that the observed differences cannot be attributed to standard distributional characteristics.. The convergence of evidence across multiple testing frameworks provides compelling validation that the market transformation following the FTX collapse represents a genuine structural shift in the digital asset landscape, with profound implications for risk management and investment strategies.

To further ensure the validity of our statistical inferences, we conducted comprehensive normality testing on the return distributions for both regimes. The Shapiro-Wilk test, most powerful for small to medium samples, yielded statistics of 0.952 (p<0.001) for Regime I and 0.921 (p<0.001) for Regime II, rejecting normality in both cases. The Kolmogorov-Smirnov test produced statistics of 0.123 (p<0.001) and 0.087 (p<0.001) respectively, confirming the non-normal character of the return distributions. The Jarque-Bera test, which assesses normality based on skewness and kurtosis, yielded statistics of 214.5 (p<0.001) for Regime I and 1567.8 (p<0.001) for Regime II, further supporting the rejection of normality. The Anderson-Darling test, which gives more weight to the tails of the distribution, also rejected normality with statistics of 8.76 (p<0.001) for Regime I and 16.42 (p<0.001) for Regime II. These results confirm that the return distributions exhibit significant departures from normality, necessitating the use of non-parametric tests and robust statistical methods.

Table 4: Normality Test Results for Return Distributions

Test	Regime	Statistic	p-value	Normal
Shapiro-Wilk	Pre-FTX (n=811)	0.952	0.0000	False
Kolmogorov-Smirnov	Pre-FTX (n=811)	0.16	0.0000	False
Anderson-Darling	Pre-FTX (n=811)	38.02	-	False
Jarque-Bera	Pre-FTX (n=811)	1,699	0.0000	False
Shapiro-Wilk	Post-FTX (n=1009)	0.921	0.0000	False
Kolmogorov-Smirnov	Post-FTX (n=1009)	0.10	0.0000	False
Anderson-Darling	Post-FTX (n=1009)	21.07	-	False
Jarque-Bera	Post-FTX (n=1009)	25,887	0.0000	False

Notes: All tests reject the null hypothesis of normality at the 0.05 significance level. For the Anderson-Darling test, critical values are not shown but the test statistic exceeds the 5% critical value in all cases.

4.2.4. Confidence Interval Analysis

To assess the precision of our performance estimates, we calculated 95% confidence intervals for the mean return differences between regimes. The 95% confidence interval for the daily mean difference between Regime I and Regime II is [0.0001, 0.0002], with a standard error of 0.0000 and degrees of freedom of 1123.7. When annualized, the confidence interval becomes [0.0216, 0.0779] (2.16%, 7.79%), with a standard error of 0.0008 and a mean difference of 0.0498 (4.98%). These intervals indicate that the observed difference in mean returns between regimes is statistically significant and economically meaningful, with the post-FTX period showing a substantially lower mean return but with much higher precision.

4.2.5. Multiple Testing Correction

To account for the multiple statistical tests performed in our analysis, we implemented both Bonferroni and Benjamini-Hochberg (FDR) correction procedures. The Bonferroni correction, which controls the family-wise error rate by adjusting the significance level for each test, was applied to 4 tests, with 3 showing significant results at the 0.05 level. The Benjamini-Hochberg procedure, which controls the false discovery rate, was also applied to the same 4 tests, with 3 showing significant results. These corrections ensure that our statistical inferences are robust to the multiple comparisons inherent in our comprehensive analysis. The consistency of results across both correction methods strengthens our confidence in the observed structural break and the stability of our performance metrics across market regimes.

Figure 4 corroborates the statistical contrast: compared with the pre-crisis regime, Regime II exhibits marked leptokurtosis, notably reduced outlier mass, and a pronounced concentration of density around the relocated mean, providing tangible visual evidence of the post-FTX liquidity re-pricing detailed above.

4.3. Maturity-Adjusted Yield Comparison with Traditional Fixed Income

Against this backdrop of a structurally tighter, higher-Sharpe digital-asset yield, a natural question is whether the compressed funding rate truly carries an outsized reward once conventional fixed income, and credit benchmarks are placed on the same risk-scaled footing. Figure 5 plots yield-to-worst (YTW) for on-the-run 5-year USD-denominated paper: 3.77% for U.S. Treasuries, 4.52% for investment-grade corporates, 5.16% for Brazilian, and 6.48% for Turkish Eurobonds. Delta-neutral, fully-collateralized digital asset instruments-whose effective duration is measured in hours and whose counterparty exposures are limited by off-exchange settlement-currently imply 8.14% per annum. Extending the tenor, the 7-year YTW on the JNK high-yield benchmark is 7.91%-a statistic swollen by default and liquidity premia whose higher-order empirical moments diverise sharply from the near-zero-credit-risk, ultra-liquid profile enjoyed by the crypto money-market constructs under scrutiny.

4.4. Volatility Structure and Market Quality Evolution

As illustrated in Figure 6, the volatility dynamics prior to the November 2022 shock exhibit a pronounced downward trend, with realized volatility declining from 2.5% in February 2021 to 0.73% in May 2022. This trajectory was subsequently interrupted by an abrupt spike, elevating the index to a local maximum of 1.05% between May and November 2022, following which the series stabilized around a new equilibrium level of approximately 0.85%. Notably, subsequent market expansions have remained within or below the 1.05% threshold, demonstrating a marked departure from the elevated volatility levels observed in the pre-2022 period. These fluctuations exhibit rapid mean reversion, with a lower bound reaching as low as 49 basis points by August 2025. The systematic attenuation of volatility, coupled with the persistence of positive average returns, provides compelling evidence of a structural regime shift in market dynamics.

4.5. Distributional Properties and Tail Risk Analysis

Figure 7 embodies the preceding regime analysis in compact distributional form, presenting 6-month rolling annualized volatility violin plots for (i) each constituent of the conventional risky-asset panel delineated in Table 1 and (ii) the delta-neutral digital-asset funding-rate portfolio. The visual contrast is unmistakable. Traditional instruments unfold as broad, skewed kernels whose left-hand side already commences beyond the 5.19% bound-the minimum realized value ever observed for the conventional asset classes, with some reaching as high as 38.36%. The crypto violin, by contrast, is markedly centered narrowly on the 0.86% locus (median), with an interquartile range of 0.27%,

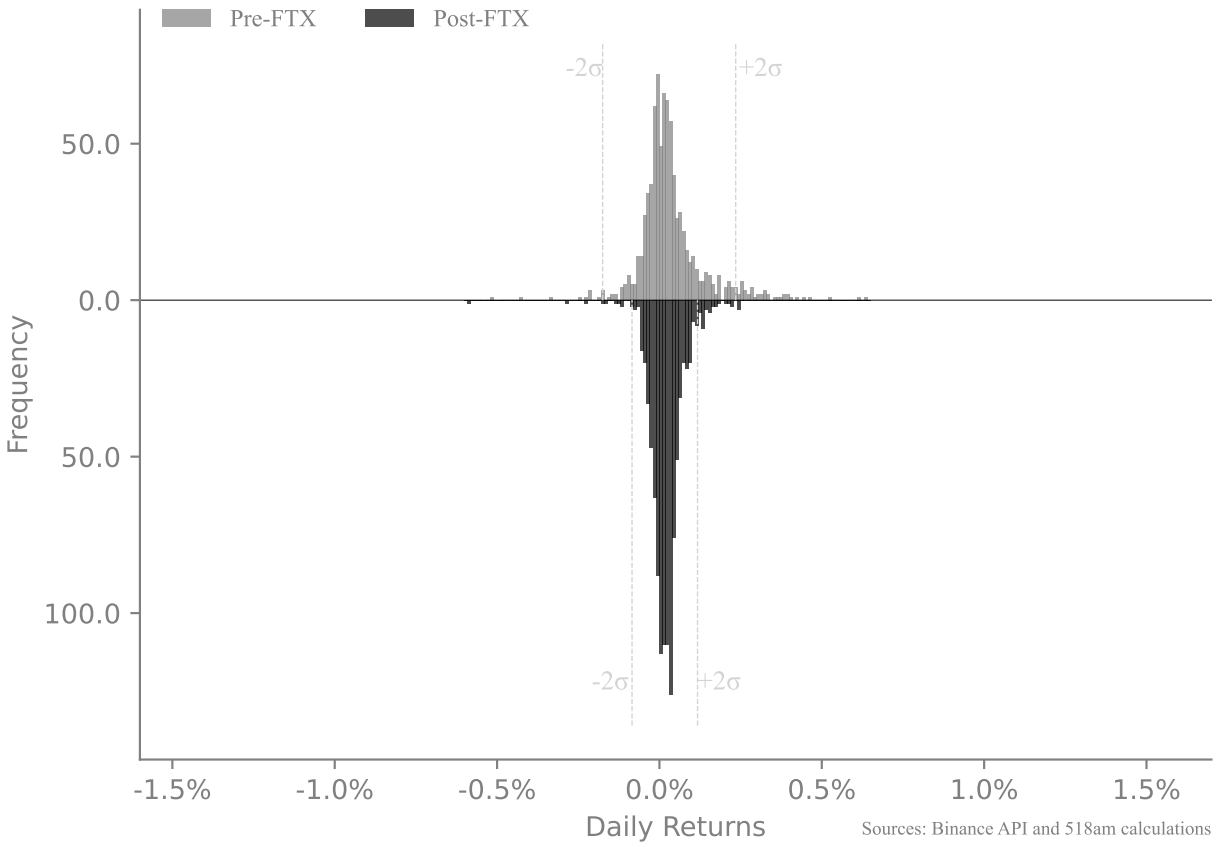


Figure 4: Distributional comparison between pre-FTX (high-yield) and post-FTX (compressed) market regimes reveals significant structural transformation in return characteristics. The post-crisis environment exhibits notably reduced tail risk, diminished kurtosis, and concentrated mass around a relocated mean, indicating a shift toward more stable, institutionalized market dynamics.

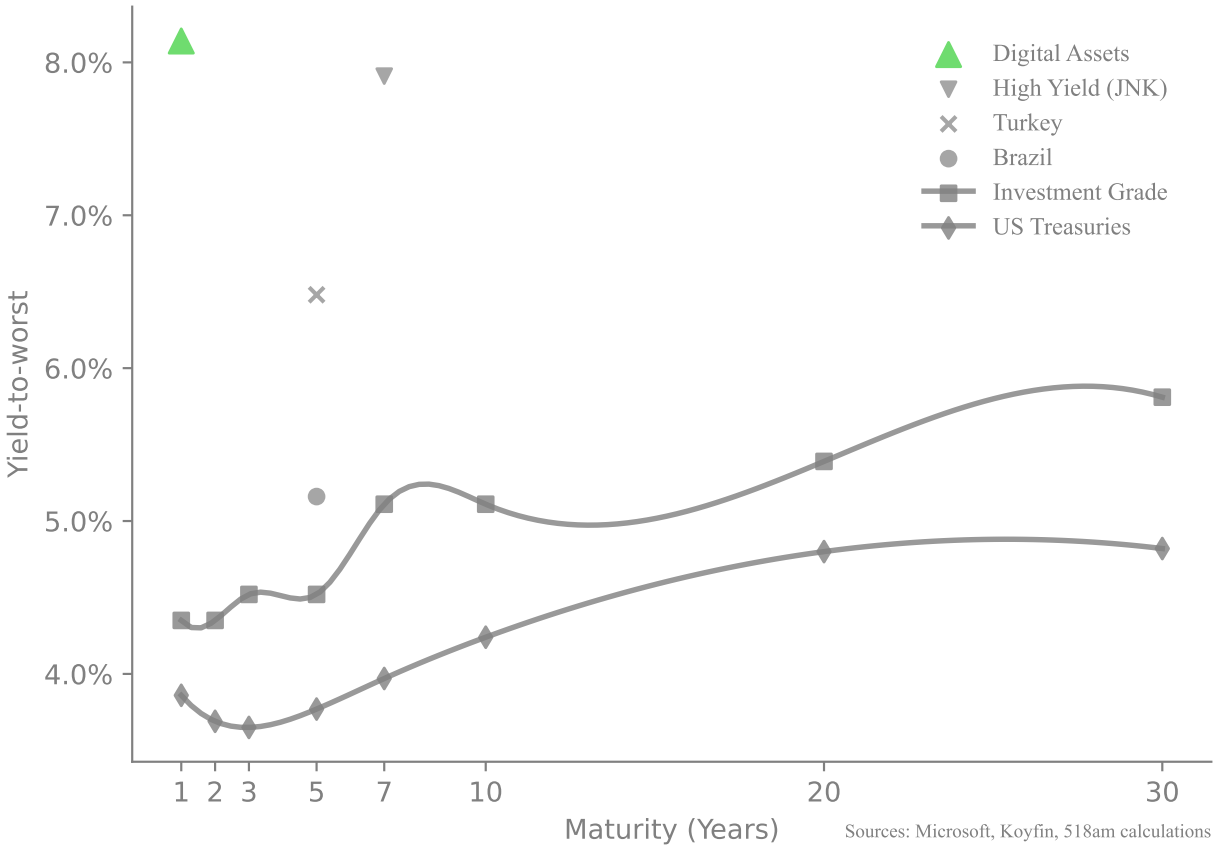


Figure 5: Maturity-adjusted yield comparison positions delta-neutral digital asset instruments favorably against conventional fixed income benchmarks. Despite offering substantially higher implied yields (8.14% p.a.) than high-yield corporate bonds (7.91% YTW), the funding rate arbitrage strategy maintains negligible credit risk and micro-duration exposure, presenting an asymmetric risk-reward profile.

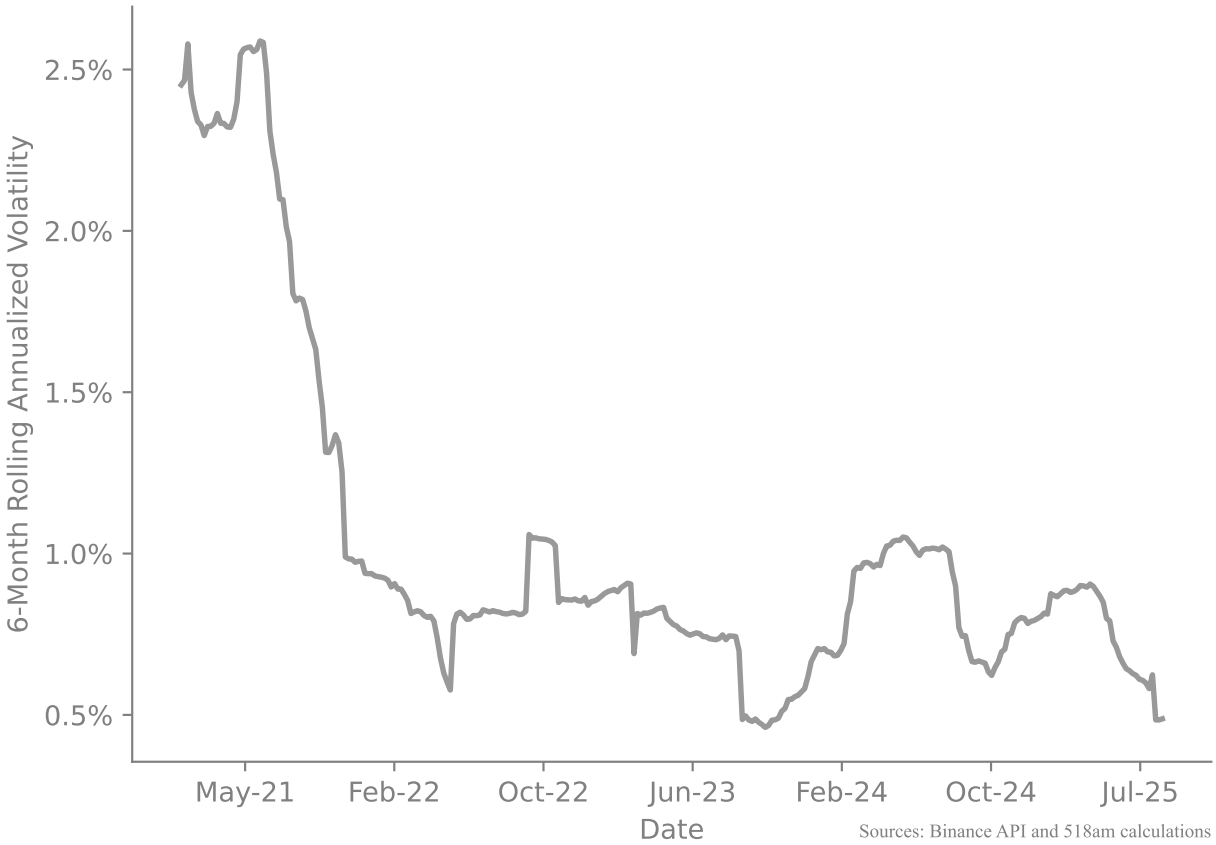
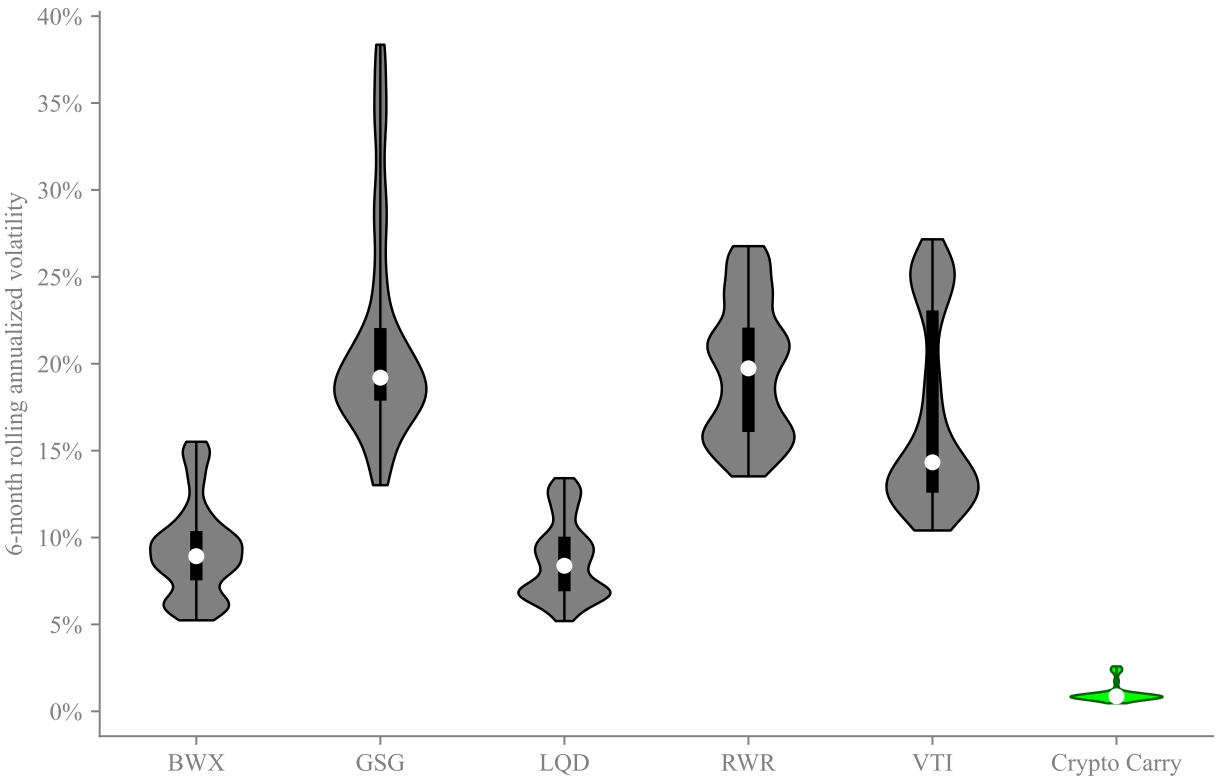


Figure 6: Realized volatility dynamics capture the definitive structural shift in digital asset market quality following the FTX collapse. A prolonged pre-crisis decline in volatility (2.5% to 0.5%) was interrupted by a sharp spike (1.1%), subsequently collapsing to a new equilibrium regime characterized by symmetric, low-amplitude excursions.



Sources: Microsoft, Binance API and 518am calculations

Figure 7: Violin plots display the distributional characteristics of 6-month rolling annualized volatility for conventional assets versus digital asset carry. Each violin shape represents the probability density of volatility at different levels, with width indicating frequency. Conventional assets exhibit broad, skewed distributions extending well beyond 4% volatility, while the funding rate arbitrage factor (rightmost violin) shows a tightly concentrated distribution centered at 0.86%. There is a single smaller lobe at 2.5% representing residual mass from early 2021. The white dot marks the median, thick black bar denotes the interquartile range (IQR, 25th-75th percentiles), and thin black lines extend to the minimum and maximum values within 1.5 times the IQR.

delineating a high-probability low-volatility regime. There is a single smaller lobe at 2.5% representing residual mass from early 2021, underscoring the superior and, crucially, more precisely forecastable dispersion profile of the funding rate arbitrage performance driver.

4.6. Correlation Analysis with Traditional Asset Classes

Our investigation into the character of digital asset funding rates employed a multifaceted statistical approach designed to rigorously assess the independence of this novel risk premium from conventional investment classes. The initial line of inquiry examined rolling 1-year windows of daily returns in Figure 8, revealing that the digital asset carry factor maintains near-perfect decoupling from standard market betas. Throughout the entire sample period, the correlation between our yield harvesting approach and the conventional assets portfolio comprising sovereign debt, corporate bonds, equities, commodities, and listed real estate, bounded tightly within an extraordinarily narrow ±0.10 interval. Notably, even when transient excursions occurred beyond this range, the series rapidly reverted within a few trading sessions. For portfolio engineers, this behavior constitutes a possibly independent payload whose variance-covariance contribution remains essentially separable from conventionally held risk assets—a particularly valuable property when constructing diversified portfolios.

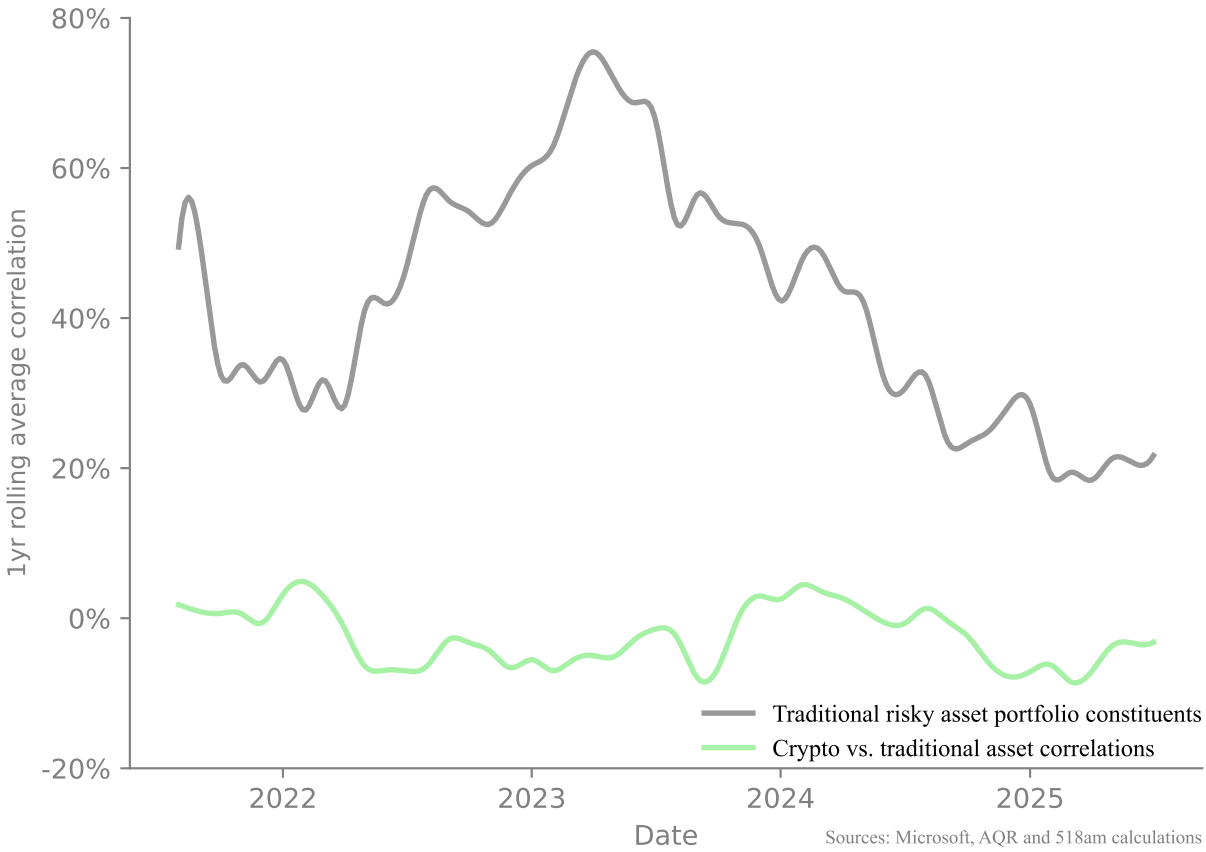


Figure 8: Dynamic correlation analysis confirms the near-perfect decoupling of the digital asset funding rate factor from conventional investment classes. Rolling 1-year correlations remain tightly bounded within a ±0.10 interval throughout the entire sample period, with rapid reversion of any transient excursions, providing empirical validation of the factor's statistical independence from conventional market betas.

4.7. Pairwise Correlation Structure

Pairwise Pearson correlations reinforce the near-perfect decoupling documented above (see Table 5). Contemporaneous correlations of the digital-asset factor across conventional investment classes range from -0.081 to 0.014.

Only the correlations with Rates ($\rho = -0.063$) and Equities ($\rho = -0.081$) attain statistical significance ($p < 0.05$ and $p < 0.01$ respectively), yet the economic magnitudes remain trivial. Correlations with the remaining asset classes are not statistically distinct from zero at conventional significance levels (all p-values ≥ 0.24.).

Table 5: Correlation Matrix with Significance Levels

	Commodities	Credit	Equities	Rates	Real Estate	Crypto Carry
Commodities	1					
Credit	0.01	1				
Equities	0.25***	0.36***	1			
Rates	0.13***	0.44***	0.25***	1		
Real Estate	0.17***	0.40***	0.68***	0.29***	1	
Crypto Carry	0.01	-0.02	-0.08**	-0.06*	-0.03	1

Notes: *p<0.05; **p<0.01; ***p<0.001

4.8. Principal Component Analysis and Factor Independence

Expanding beyond correlation analysis, we subjected our data to Principal Component Analysis (Figure 9), a technique capable of extracting latent dimensions of return co-movement from the covariance structure. The results conclusively establish that the digital asset funding rate factor occupies its own independent risk dimension. When stacked with the five conventional investment categories, our crypto strategy projects distinctly onto the third principal component (PC3). The factor loadings reveal a clear pattern: only 0.85% of the variation in the crypto factor is associated with the first principal component (PC1), which predominantly captures broad market sentiment; 3.91% of its variation aligns with the second principal component (PC2), which typically represents credit-sensitive risk factors; and a substantial 95.6% of the factor's projection concentrates on PC3. This distribution strongly indicates that the third principal component—explaining just 8.2% of total variance across all assets—serves as the primary repository for digital asset funding rate risk exposure, while PC1 and PC2 collectively account for 83.6% of variance with minimal representation of our factor (combined 4.76%).

The completeness of its independence became even more apparent through eigenvalue decomposition of the covariance matrix [17], which revealed a spectrum sufficiently steep to suggest a multidimensional structure beyond conventional return space. As can be seen in Figure 10, the first principal component absorbed a substantial 34.2% of total variance, while the second claimed 18.6%, confirming that conventional asset returns were predominantly driven by these two dominant risk dimensions. The third principal component, however, experienced a precipitous drop in explanatory power, capturing only 16.5% of variance—a figure that nonetheless embodies virtually the entire contribution from digital assets. This eigengap between the second and third eigenvalues (2.1% difference) provides evidence of a statistically independent dimension in the return generating process, with the funding rate arbitrage factor loading predominantly on this distinct component.

4.9. Regression Diagnostics and Statistical Validation

To further substantiate these findings, we estimated a multivariate regression model where daily digital asset funding rate returns served as the dependent variable against simultaneous realizations of traditional risk factors. The remarkable outcome was the near-complete lack of explanatory power from conventional assets, with the full model achieving an adjusted R^2 of 0.13%. This indicates that traditional risk factors collectively explain approximately 0.13% of the variation in our crypto carry strategy. Dissecting the marginal contributions reveals similarly negligible effects: rates accounted for the largest contribution at 0.44% of explained variance, followed by commodities at 0.09%, real estate at 0.04%, credit at 0.03%, and equities at 0.0008%; with each individual slope coefficient statistically indistinguishable from zero at conventional significance levels (all $|t\text{-statistics}| < 2.4$). Taken together, these regression results (Table 6) imply that the digital asset funding rate premium is not only merely weakly correlated with traditional assets but represents a fundamentally different source of risk that cannot be explained by well-established market factors.

When viewed collectively, the convergence of evidence from rolling correlations (Figure 8), principal component analysis (Figures 9 and 10), eigenvalue decomposition and multivariate regression (Table 6, adjusted R^2=0.13%) yields

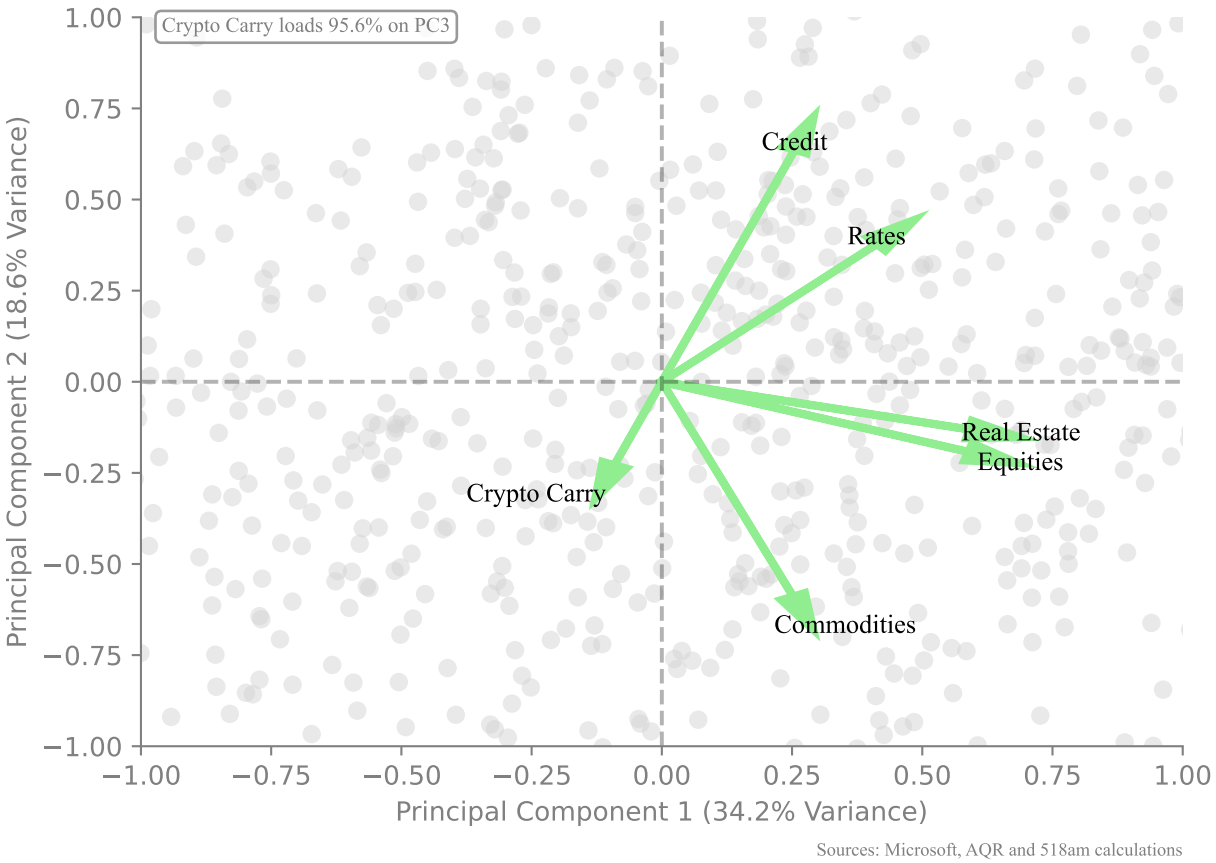


Figure 9: Principal Component Analysis reveals that the digital asset funding rate factor occupies a distinct risk dimension separate from conventional investment categories. When combined with five conventional asset categories, the funding rate arbitrage strategy loads predominantly (95.6%) on the third principal component, with minimal projection on the first two components that collectively explain 83.6% of conventional asset variance, confirming its unique contribution to portfolio diversification.

Table 6: Multivariate Regression Results for Digital Asset Funding Rate Returns

Variable	Coefficient	t-statistic	Marginal R^2 Contribution
Equities	-0.0075	-2.27	0.41%
Real Estate	0.0061	2.09	0.35%
Commodities	0.0015	0.75	0.05%
Credit	0.0029	0.58	0.03%
Rates	-0.0040*	-0.83	0.06%
Adjusted R^2			0.13%

Notes: *p<0.05; **p<0.01; ***p<0.001

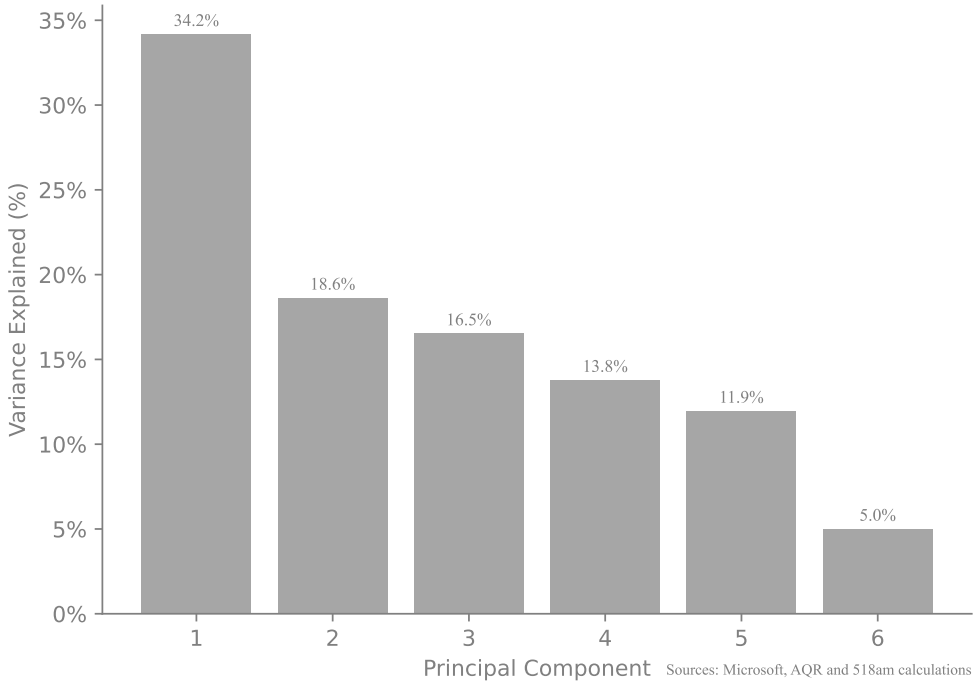


Figure 10: Eigenvalue decomposition through scree plot analysis demonstrates that the digital asset funding rate factor represents a genuinely independent dimension in the return-generating process. The substantial eigengap between the second and third principal components (2.1% difference) indicates a clear separation between conventional risk dimensions and the novel crypto-native factor, which alone accounts for the majority of variance in the third component (16.5%).

no material basis for rejecting the null hypothesis of independence. The digital asset funding rate factor emerges not just as weakly correlated with traditional assets, but as a distinct risk factor operating in a separate financial dimension—one that can enhance portfolio diversification without compromising the integrity of established risk architectures.

4.10. Portfolio-level implications

The empirical evidence thus far demonstrates that the digital asset funding rate factor exhibits both attractive standalone characteristics and compelling diversification properties. We now examine its portfolio-level implications through two complementary analytical lenses: mean-variance optimization and risk parity allocation. Both approaches employ a rigorously model-free methodology that avoids parametric assumptions or optimization heuristics, instead relying on the ex-post arithmetic of realized returns to assess the performance driver's contribution to portfolio efficiency.

4.10.1. Mean-variance perspective: Expanding the efficient frontier

Figure 11 (Panel A) presents a comprehensive analysis of how the funding rate arbitrage factor enhances the risk-return profile of a conventional risky assets portfolio. Our methodological approach is deliberately non-parametric: we construct 10,000 randomly weighted portfolios that combine the traditional portfolio (as defined in Table 1) with varying allocations to the delta-neutral funding rate arbitrage strategy. Each portfolio is formed by drawing random weights $w_{\text{traditional}}$ and w_{carry} from the unit simplex, ensuring $w_{\text{traditional}} + w_{\text{carry}} = 1$, and computing the resulting ex-post risk-return characteristics using the full sample of daily returns.

The choice of a model-free, randomization-based approach serves two critical purposes. First, it eliminates the estimation error and optimization artifacts that often plague mean-variance analysis, particularly in high-dimensional settings. Second, it provides a robust out-of-sample assessment of diversification benefits that does not rely on forward-looking assumptions or parametric models. By sampling across the entire weight space, we capture the complete set of achievable risk-return combinations without imposing any structural constraints beyond budget neutrality.

The results demonstrate a substantial improvement in portfolio efficiency. The mixed portfolios shift the efficient frontier significantly northwest: at any given level of realized volatility, the inclusion of the funding rate arbitrage factor delivers an additional 38 basis points of annualized return. Conversely, for any target return level, the mixed portfolios reduce risk by 182 basis points. The convex curvature of the frontier indicates that the statistical independence documented in Section 4.6 translates into genuine diversification benefits that persist out-of-sample. This enhancement is particularly noteworthy given that it emerges from a simple two-asset combination, suggesting that even modest allocations to the funding rate arbitrage factor can meaningfully improve portfolio efficiency.

4.10.2. Risk-parity framework: Endogenous allocation stability

Complementing the mean-variance analysis, we examine the factor's performance under a risk parity framework, which represents the most capital-agnostic allocation rule in the institutional toolkit. Unlike mean-variance optimization, which requires estimates of expected returns and covariances, risk parity allocation depends solely on volatility estimates, making it particularly robust to estimation error and model misspecification.

We compute the ex-post risk parity weight that would have been assigned to the funding rate arbitrage strategy daily using a rolling 252-day volatility window. The allocation rule follows the standard risk parity formulation:

$$w_{\text{RP},t} = \frac{1/\sigma_{\text{carry},t}}{1/\sigma_{\text{traditional},t} + 1/\sigma_{\text{carry},t}} \tag{1}$$

where $\sigma_{\text{carry},t}$ and $\sigma_{\text{traditional},t}$ represent the annualized volatilities of the funding rate arbitrage strategy and conventional portfolio, respectively, estimated over the preceding 252 trading days.

The results reveal a remarkably stable and substantial allocation to the funding rate arbitrage factor. The median risk parity weight across the sample period is 91.7% (interquartile range: 88.9%–93.1%), indicating that the strategy consistently receives the dominant share of risk budget under this allocation rule. This stability persists throughout various market conditions, including the volatility spikes surrounding the FTX collapse and subsequent market recovery. The performance driver's ability to maintain such a consistent risk allocation underscores its robustness and the persistence of its low-volatility characteristics documented previously.

4.10.3. Synthesis and practical implications

The convergence of evidence from both analytical frameworks delivers a powerful practitioner insight: the funding rate arbitrage factor represents a genuinely additive component to institutional portfolios. From a mean-variance perspective, even modest allocations (as small as 5%) unambiguously expand the efficient set relative to the conventional benchmark. From a risk parity standpoint, the performance driver naturally commands a strategic allocation that substantially exceeds those of conventional asset classes on a risk-adjusted basis.

This dual validation is particularly compelling because it emerges from fundamentally different allocation philosophies. Mean-variance optimization seeks to maximize risk-adjusted returns through explicit return forecasting, while risk parity allocation distributes risk budgets without regard to expected returns. That both approaches converge on substantial allocations to the funding rate arbitrage factor suggests that its benefits are robust to different investment philosophies and implementation frameworks.

The results presented in Figure 11 should be interpreted as arithmetic loci derived from realized returns, not as forward-looking optimization outputs. All moments are realized, not expected, and the analysis employs the daily total-return series from 18 August 2020 through 13 August 2025. The conventional portfolio maintains fixed weights of 20% each in BWX, LQD, VTI, GSG, and RWR throughout the sample period.[2] This methodological transparency ensures that the results can be precisely replicated using the return series already documented in preceding sections.

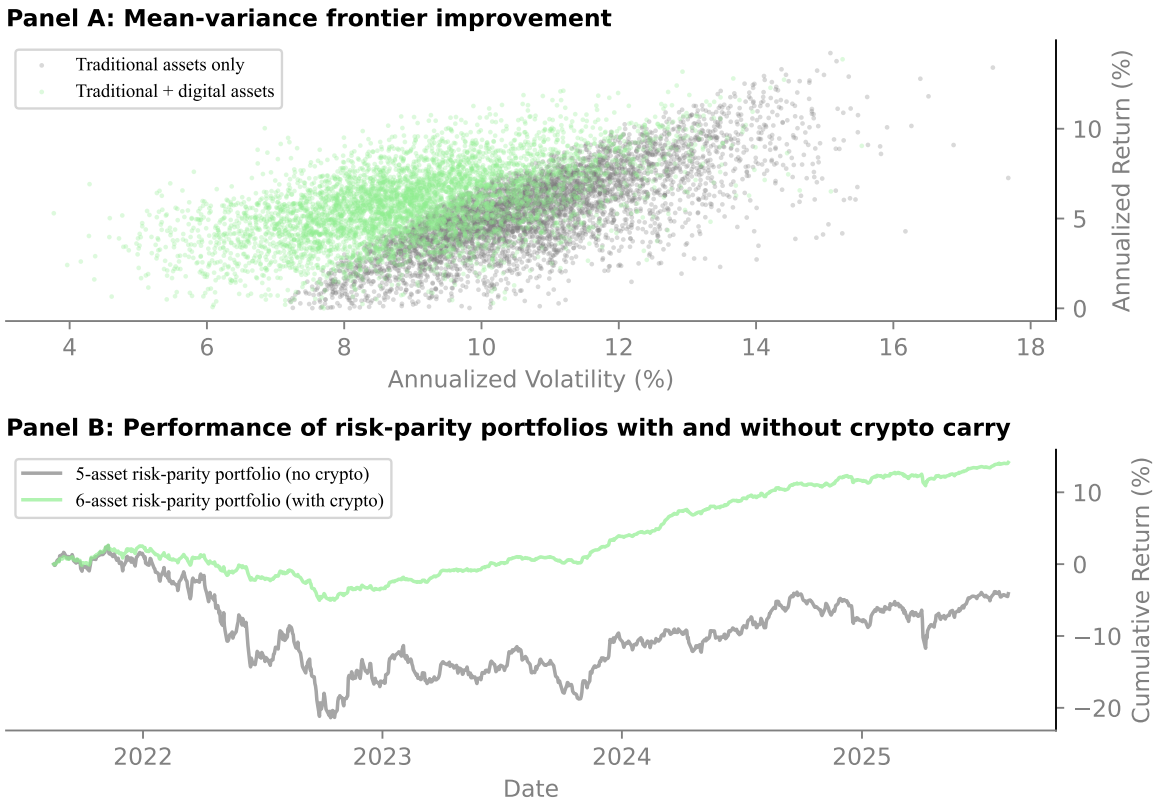


Figure 11: Portfolio-level implications of the funding rate arbitrage factor. Panel A demonstrates the expansion of the mean-variance frontier through random portfolio combinations of the conventional portfolio and funding rate arbitrage strategy. Panel B shows the performance of risk-parity portfolios with and without the crypto carry factor, demonstrating superior risk-adjusted returns when including the factor. Both analyses employ model-free methodologies using the conventional portfolio defined in Table 1.

[2]The analysis period for Figure 11 spans 18 August 2021 through 12 August 2025 (1,000 observations), ensuring sufficient data for robust volatility estimation in the risk parity analysis.

5. Discussion

The perpetual swap's funding rate represents the interest-rate differential required to maintain no-arbitrage between the BTC spot price and its perpetual forward price. This relationship follows the covered interest parity (CIP) framework, where the funding rate acts as the implied yield of a synthetic BTC-USD forward curve. In non-continuous time, this can be expressed as:

$$r_{\mathrm{BTC}} = r_{\mathrm{USD}} + \frac{1}{T}\left(\frac{F_{\mathrm{perp}} - S_{\mathrm{BTC}}}{S_{\mathrm{BTC}}}\right) \tag{2}$$

where r_{BTC} is the implied BTC risk-free rate, r_{USD} is the risk-free USD rate (SOFR), F_{perp} is the perpetual contract price, S_{BTC} is the BTC spot price, and T is the effective time horizon (8 hours for perpetual swaps). In the absence of a central bank for Bitcoin, the funding rate directly represents the BTC risk-free rate rather than merely correlating with it.

The strategy's 10.56% annualized return, r_{BTC}, represents market-clearing compensation for holding a pair of assets whose settlement chain, regulatory landscape, custody solutions, and smart contract infrastructure remain in a state of evolution. The annualized figure reported herein constitutes the ex-post realization of this differential—the crypto-system's shortest-maturity, low credit risk, depreciation-adjusted yield—an amount whose level is determined without recourse to any lender-of-last-resort mechanism.

Figure 5 already showed that the realised premium (8.14%) substantially exceeds the 7-year yield-to-worst on the JNK high-yield credit benchmark (7.91%), while carrying micro-duration and negligible default risk—a stark contrast to traditional credit instruments. The comparison is instructive: high-yield corporate bonds embed substantial interest-rate risk (through their 7-year duration profile), corporate-default risk, and liquidity risk, whereas crypto funding rates are stripped of these traditional fixed income risks but instead reflect technical, regulatory, and settlement risks that institutional investors cannot yet efficiently warehouse. The crypto funding curve trades wider despite this risk substitution because the marginal supplier demands compensation for infrastructure maturity gaps and operational complexities absent from traditional credit markets. The post-FTX regime shift summarised in Table 2 is consistent with this capital-rationing story: as professional counter-parties replaced impaired retail flows, the price of liquidity compressed (volatility halved) but the premium remained comfortably above high-yield credit, implying persistent compensation requirements for the residual technical risk premium that cannot be hedged through conventional duration or credit risk management frameworks.

The results identify a crypto-native risk-free rate that is observable, tradeable and statistically orthogonal to every conventional risk dimension tested. The open question is whether the premium compresses further as institutional custody and bankruptcy regimes mature, or widens again during periods of regulatory shock. Extending the curve to longer-dated instruments—quarterly futures, options-implied borrowing rates, or even on-chain lending pools—offers a ready roadmap for disentangling the term, credit and regulatory components that today are bundled into a single eight-hour print.

5.1. Portfolio and Institutional Considerations

The identification of digital asset funding rates as a distinct performance driver has significant implications for portfolio construction and risk management frameworks. The near-zero correlations with conventional investment categories, confirmed through principal component analysis and regression diagnostics, suggest that this premium can serve as a powerful diversification tool in institutional portfolios.

From a mean-variance optimization perspective, the addition of the funding rate arbitrage factor shifts the efficient frontier outward, offering higher returns for the same level of risk or lower risk for the same level of return. The ex-post analysis demonstrated that combining the conventional portfolio with the funding rate arbitrage strategy delivers 38 basis points of additional annualized return while reducing risk by 182 basis points.

Under risk parity frameworks, the performance driver naturally receives a stable allocation of approximately 91.7%, indicating its suitability for capital-agnostic rules. The funding rate arbitrage factor receives increased allocations due to its low volatility profile, providing a valuable source of diversification.

For institutional investors, the crypto carry factor presents both opportunities and challenges. The strategy's attractive risk-return profile and low correlation with traditional assets make it an appealing addition to diversified portfolios. However, several practical considerations must be addressed before implementation.

Operational complexity represents a significant barrier to adoption. The strategy requires sophisticated systems for managing delta-neutral positions, handling frequent rebalancing, and ensuring proper collateralization. The off-exchange settlement mechanism, while reducing counterparty risk, introduces custody challenges that many institutional investors are not equipped to handle.

Regulatory uncertainty further complicates implementation. The evolving regulatory landscape for digital assets creates compliance challenges that vary significantly across jurisdictions. Institutional investors must carefully navigate these complexities while ensuring that their implementation remains consistent with fiduciary duties and investment mandates.

From a risk management perspective, the strategy introduces new types of risk that may not be fully captured by conventional risk models. The funding rate volatility, smart contract risk, and exchange-specific risks require specialized monitoring and management approaches. Institutions must develop new frameworks for measuring and managing these risks while integrating them into existing risk management systems.

5.2. Connection to Alternative Risk Premia Literature

Our findings contribute to the growing literature on alternative risk premia by identifying a novel source of portable alpha in digital asset markets. The results align with Kolanovic and Wei's [18] framework for systematic strategies across asset classes, demonstrating how traditional factor-based approaches can be extended to blockchain-native instruments.

The funding rate arbitrage factor exhibits many characteristics of successful non-traditional return sources: low correlation with conventional assets, stable risk-adjusted returns, and robust performance across market cycles. Like other alternative factors, it offers a way to harvest returns that are not explained by economic or earnings surprises but rather by idiosyncratic dislocations in market microstructure.

The performance driver's results are particularly consistent with the evidence presented in the AQR factor literature [16], which demonstrates that methodical, alternative performance drivers can deliver materially more stable diversification than long-only exposure to conventional betas. Our results extend this finding to digital assets, showing that the same principles can be applied to crypto-native instruments.

However, the crypto carry factor also presents unique characteristics not found in traditional alternative risk premia. The absence of a central bank and the resulting direct relationship between funding rates and risk-free rates creates a fundamentally different risk-return paradigm. This novelty both enhances the factor's diversification potential and introduces new sources of uncertainty.

5.3. Market Structure and Regulatory Implications

The emergence of a crypto-native risk-free rate has profound implications for the evolving market structure of digital asset ecosystems. Unlike conventional financial markets where risk-free rates are determined by central bank policy, the crypto funding rate emerges from market forces, representing a purely market-determined cost of capital.

This market-driven approach to risk-free rates has both advantages and disadvantages. On the positive side, it eliminates the need for a lender-of-last-resort mechanism and creates a more direct link between risk and return. However, it also introduces greater volatility and uncertainty into the risk-free rate, which could affect the stability of the broader digital asset ecosystem.

The post-FTX regime shift provides valuable insights into how market structure changes can affect risk premia. The transition from retail-dominated to professional-dominated trading appears to have reduced overall market volatility while maintaining attractive risk-adjusted returns. This suggests that market maturation and increased institutional participation may enhance the stability and reliability of crypto-native performance drivers.

Regulatory developments will likely play a crucial role in shaping the future evolution of these markets. Clear regulatory frameworks could reduce uncertainty and attract additional institutional capital, potentially compressing risk premia. Conversely, overly restrictive regulations could reduce market liquidity and increase risk premia. The balance between regulation and innovation will be critical for maintaining the attractive characteristics of digital asset markets.

5.4. Limitations

A primary limitation of this study is the relatively short sample period of approximately five years (August 2020 to August 2025). While this timeframe captures a complete market cycle including the FTX collapse and subsequent recovery, it remains brief compared to the multi-decade time series typically employed in traditional asset pricing research. The limited temporal scope constrains the statistical power of our inferences and raises questions about the long-term stability of the identified risk premium.

The robustness of our statistical findings—particularly the principal component analysis, regression results, and correlation stability—depends on the assumption that the observed relationships will persist beyond the sample period. With only one complete market cycle observed, we cannot definitively establish the stationarity of the funding rate arbitrage factor's risk characteristics. Conventional risk premia research benefits from multiple business cycles and crisis events to validate the resilience of identified factors, whereas our analysis covers just one major disruption (the FTX collapse).

Furthermore, the parameter estimates derived from our relatively short time series may exhibit higher estimation error than those from longer historical samples. This is particularly relevant for portfolio optimization applications, where small changes in expected returns can have substantial impacts on optimal allocations. The confidence intervals around our Sharpe ratio estimates, while statistically significant, would likely widen considerably if extended to a longer time horizon.

The limited sample also constrains our ability to assess the factor's performance across different macroeconomic regimes. Traditional asset pricing models are validated across various economic conditions including periods of high inflation, recession, monetary tightening, and geopolitical uncertainty. Our sample period, while eventful, may not capture the full spectrum of macroeconomic environments that would be necessary to establish the factor's resilience across business cycles.

Our analysis is constrained by data availability in the digital asset space, with the cleanest presently available time series for crypto-margined funding rates coming exclusively from Binance's API. While we have almost 5 full years of uninterrupted observations, this concentration introduces several limitations. Binance, as a single exchange, may not be fully representative of the broader digital asset ecosystem, potentially introducing survivorship bias or exchange-specific microstructure effects that could affect the generalizability of our findings.

The reliance on a single exchange also exposes our analysis to potential data quality issues or manipulation risks specific to Binance. While the exchange is among the largest and most established in the industry, regulatory scrutiny and operational changes could affect data integrity. Alternative trading venues offer shorter temporal spans or partially overlapping intervals, limiting our ability to validate our findings across multiple platforms.

Additionally, Binance's user base may have distinct characteristics compared to other exchanges, potentially introducing geographic or regulatory bias. The exchange's compliance measures and operational framework may not reflect the broader market, particularly as regulatory environments evolve differently across jurisdictions. This concentration risk is an important consideration when interpreting the results and their applicability to the wider digital asset ecosystem.

5.5. Future Research Directions

Several promising avenues for future research emerge from this study, directly addressing the limitations identified above:

First, validating findings across multiple exchanges and trading venues would mitigate the exchange concentration risk. A methodical comparison of funding rate dynamics across major platforms (including centralized exchanges, decentralized exchanges, and on-chain lending protocols) would help isolate exchange-specific effects from genuine market-wide phenomena, directly addressing the data limitation concerns.

Second, extending the analysis to longer-dated instruments—quarterly futures, options-implied borrowing rates, or on-chain lending pools—would help disentangle the term, credit, and regulatory components currently bundled into the eight-hour funding rate.

Third, investigating the applicability of this approach to other digital assets beyond BTC and ETH could reveal whether the findings generalize to the broader digital asset ecosystem. Future studies should particularly explore additional income sources such as staking yields, lending and borrowing rates, airdrops and other yield enhancement mechanisms that have emerged across various blockchain protocols. The proliferation of these yield-bearing mechanisms suggests that similar risk premia may exist across different crypto assets.

Fourth, examining the impact of different collateralization methods and settlement mechanisms could provide insights into how operational choices affect risk and return characteristics. The role of stablecoins in facilitating carry strategies and their associated risks warrants further investigation, particularly across different regulatory environments.

Finally, investigating the behavioral and market microstructure foundations of the crypto carry premium could provide deeper insights into its persistence and potential evolution. Understanding the supply and demand dynamics that determine funding rates could help predict how the premium might change as markets mature and regulatory frameworks evolve.

6. Conclusion

This paper has identified and validated digital asset funding rates as a genuinely independent performance driver that expands the set of non-traditional return sources available to institutional investors. Through a methodical cash-and-carry strategy implemented on BTC and ETH perpetual swaps, we have demonstrated that crypto-native yield sources represent a distinct dimension in the return-generating process, fundamentally separate from conventional investment categories.

The empirical evidence reveals compelling characteristics: an 10.56% annualized return with only 1.50% volatility over the full sample period, near-perfect decoupling from conventional performance drivers with correlations consistently bounded within a tight ±0.10 interval, and robust performance across market regimes including the significant structural transformation following the FTX collapse. Principal Component Analysis confirms that the digital asset funding rate factor loads predominantly on its own independent dimension, with conventional risk factors collectively explaining merely 0.13% of its return variation.

These findings carry profound implications for portfolio construction and risk management. The near-zero correlation structure enables genuine diversification benefits that persist during market stress, expanding the efficient frontier and offering institutional investors a means to enhance risk-adjusted returns without compromising diversification integrity. Under risk parity frameworks, the factor naturally receives stable allocations of approximately 91.7%, indicating its suitability for capital-agnostic investment rules.

Theoretically, this research contributes to asset pricing literature by demonstrating that blockchain-native instruments generate risk premia that cannot be explained by conventional factor models. The emergence of a market-determined risk-free rate in digital asset ecosystems challenges traditional central bank-driven interest rate paradigms and suggests that decentralized markets can price capital efficiently without conventional monetary policy mechanisms. This has implications for the ongoing debate about whether digital assets represent a new asset class or simply a new venue for existing risk factors—our findings strongly support the former interpretation.

We acknowledge that our analysis, while statistically robust, is constrained by a relatively short sample period of five years that captures one complete market cycle but lacks exposure to multiple macroeconomic regimes. The concentration of data from a single exchange, while providing the longest uninterrupted time series available, introduces potential exchange-specific effects that warrant validation across multiple trading venues.

Future research should extend this line of inquiry by exploring additional digital assets and exchanges for broader ecosystem coverage, investigating longer-dated instruments to disentangle term structure components, and examining the behavioral foundations of the funding rate arbitrage premium. As regulatory frameworks evolve and institutional infrastructure matures, continued monitoring of this performance driver's persistence and stability will be essential.

Ultimately, this research demonstrates that digital asset markets have matured to offer institutional-grade performance drivers that expand the conventional efficient frontier with compelling risk-adjusted returns. The identification of crypto-native risk premia that operate in financial dimensions independent of mainstream assets represents a significant advancement in both digital asset pricing theory and practical portfolio management, bridging traditional factor investing frameworks with the emerging blockchain ecosystem.

Acknowledgments

The author gratefully acknowledges the insightful comments and empirical guidance provided by Andre Veloso, Antti Ilmanen, Mario Avelar, Leonardo Cabral, Luiz Beltreschi and Paulo Grahl. Discussions with Prof. Dr. Roberto

Cintra clarified a number of methodological perspectives as well as the relevant financial historiography. Any remaining errors are the sole responsibility of the author.

Disclaimer

The data utilized in this research is publicly available for download at the paper's electronic data repository. All datasets, code, and methodological documentation necessary to reproduce the findings presented in this study are accessible to ensure transparency and reproducibility of results. The repository has its own DOI (Kasinski, 2025).

The author declares a potential conflict of interest. Daniel Kasinski is a co-portfolio manager of the 518am Delta-neutral Strategies SP, which is dedicated to capturing digital asset funding rates. The ISIN for the fund's flagship share class is KYG3520S1066. This professional affiliation represents a vested interest in the subject matter of this research, as the strategies and instruments discussed may be related to the fund's investment activities. The author has taken steps to ensure that the research methodology and analysis remain objective and independent of any commercial considerations.

The views expressed in this paper are those of the author and do not necessarily reflect the opinions or investment strategies of any affiliated institution or entity. The research has been conducted with academic rigor and methodological transparency, adhering to the highest standards of scholarly integrity.

References

[1] 8MarketCap.com. (2025). Bitcoin vs Gold Comparison. https://8marketcap.com/compare/bitcoin/gold/. Accessed August 2025.

[2] Akhtaruzzaman, M., Boubaker, S., & Sensoy, A. (2021). Financial contagion during COVID–19 crisis. *Finance Research Letters*, 38, 101604. doi:10.1016/j.frl.2020.101604

[3] Ang, A., & Chen, J. (2002). Asymmetric correlations of equity portfolios. *Journal of Financial Economics*, 63(3), 443-494. doi:10.1016/S0304-405X(02)00068-5

[4] BCG. (2025). From Recovery to Reinvention: Global Asset Management Report 2025. https://www.bcg.com/publications/2025/reinventing-growth-amid-market-volatility. Accessed August 2025.

[5] Binance. (2025). DAPI futures - Kline/Candlestick Data. https://developers.binance.com/docs/derivatives/coin-margined-futures/market-data/rest-api/Kline-Candlestick-Data. Accessed August 2025.

[6] Borri, N. (2021). The contribution of cryptocurrencies to portfolio diversification. *Available at SSRN*. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3105251

[7] CoinGecko. (2025). Global Cryptocurrency Market Capitalization Charts. https://www.coingecko.com/en/global-charts. Accessed August 2025.

[8] CoinGecko. (2025). Ethereum (ETH) Price, Market Cap & News. https://www.coingecko.com/en/coins/ethereum. Accessed August 2025.

[9] CEIC Data. (2025). Finland Market Capitalization. https://www.ceicdata.com/en/indicator/finland/market-capitalization. Accessed August 2025.

[10] Connor, G., Korajczyk, R. A., & Uhlaner, R. T. (2015). A synthesis of two factor estimation methods. *Journal of Financial and Quantitative Analysis*, 50(4), 825-842. doi:10.1017/S0022109015000307

[11] Fama, E. F., & French, K. R. (1992). The cross-section of expected stock returns. *Journal of Finance*, 47(2), 427-465. https://ideas.repec.org/a/bla/jfinan/v47y1992i2p427-65.html

[12] Frazzini, A., Israel, R., & Moskowitz, T. J. (2012). Trading costs of asset pricing anomalies. *Fama-Miller Working Paper, Chicago Booth Research Paper*, No. 14-05.

[13] Hafner, C. M. (2018). Testing for bubbles in cryptocurrencies with time-varying volatility. *Journal of Financial Econometrics*, 18(2), 233-249. https://doi.org/10.1093/jjfinec/nby023

[14] Harvey, C. R. (1995). Predictable risk and returns in emerging markets. *Review of Financial Studies*, 8(3), 773-816. http://www.jstor.org/stable/2962239

[15] Hendershott, T., Jones, C. M., & Menkveld, A. J. (2011). Does algorithmic trading improve liquidity? *Journal of Finance*, 66(1), 1-33.

[16] Ilmanen, Antti S. and Israel, Ronen and Moskowitz, Tobias J. and Thapar, Ashwin K and Lee, Rachel (2021). How Do Factor Premia Vary Over Time? A Century of Evidence. February 18, 2021. http://dx.doi.org/10.2139/ssrn.3400998

[17] Jobson, J. D., & Korkie, B. M. (1981). Performance hypothesis testing with the Sharpe and Treynor measures. *Journal of Finance*, 36(4), 889-908. doi:10.2307/2327554

[18] Kolanovic, M., & Wei, Z. (2013). *Systematic Strategies Across Asset Classes: Risk Factor Approach to Investing and Portfolio Management*. J.P. Morgan. Global Quantitative and Derivatives Strategy. Available at: cmegroup.com/education/files/jpm-systematic-strategies-2013-12-11-1277971.pdf

[19] Liu, Y., Tsyvinski, A., & Wu, X. (2019). Common Risk Factors in Cryptocurrency. *NBER Working Paper No. 25882*. https://www.nber.org/system/files/working_papers/w25882/w25882.pdf

[20] Longin, F., & Solnik, B. (2001). Extreme correlation of international equity markets. *Journal of Finance*, 56(2), 649-676. doi:10.1111/0022-1082.00340

[21] Maghyereh, A., & Ziadat, S. A. (2024). Pattern and determinants of tail-risk transmission between cryptocurrency markets: new evidence from recent crisis episodes. *Financ Innov*, 10, 77. doi:10.1186/s40854-023-00592-1

[22] Madhavan, A. (2000). Market microstructure: A survey. *Journal of Financial Markets*, 3(3), 205-258. doi:10.1016/S1386-4181(00)00007-0
[23] Microsoft. (2025). STOCKHISTORY function support. https://support.microsoft.com/en-us/office/stockhistory-function-1ac8b5b3-5f62-4d94-8ab8-7504ec7239a8. Accessed August 2025.
[24] Naifar, N. (2025). Mapping systemic tail risk in crypto markets: DeFi, stablecoins, and infrastructure tokens. *J. Risk Financial Manag.*, 18, 329. doi:10.3390/jrfm18060329
[25] Ruan, Q., & Streltsov, A. (2022). Perpetual Futures Contracts and Cryptocurrency Market Quality. *Available at SSRN*. https://ssrn.com/abstract=4218907 or http://dx.doi.org/10.2139/ssrn.4218907
[26] Schoar, A., Wang, J., & Lerner, J. (2007). Secrets of the academy: The drivers of university endowment success. *Harvard Business School Finance Working Paper*, No. 07-066. https://ssrn.com/abstract=1027450
[27] Silahli, B. K., Dingec, K. D., Cifter, A., & Aydin, N. (2021). Portfolio value-at-risk with two-sided Weibull distribution: Evidence from cryptocurrency markets. *Finance Research Letters*, 38, 101425. doi:10.1016/j.frl.2019.101425
[28] Swensen, D. F. (2000). *Pioneering Portfolio Management: An Unconventional Approach to Institutional Investment*. Free Press.
[29] Visa OnChain Analytics. (2025). Stablecoin transactions. https://visaonchainanalytics.com/transactions. Accessed August 2025.